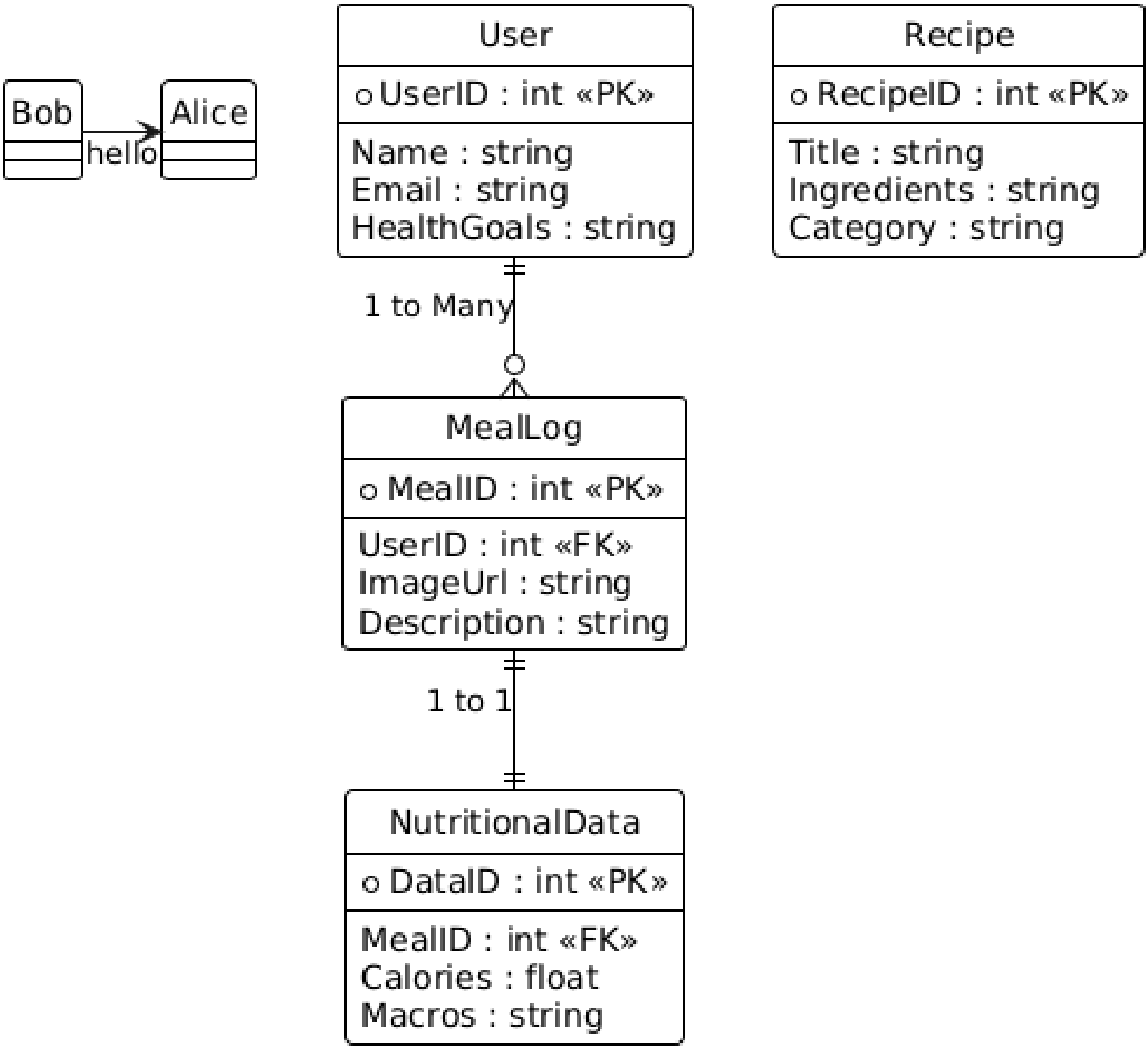
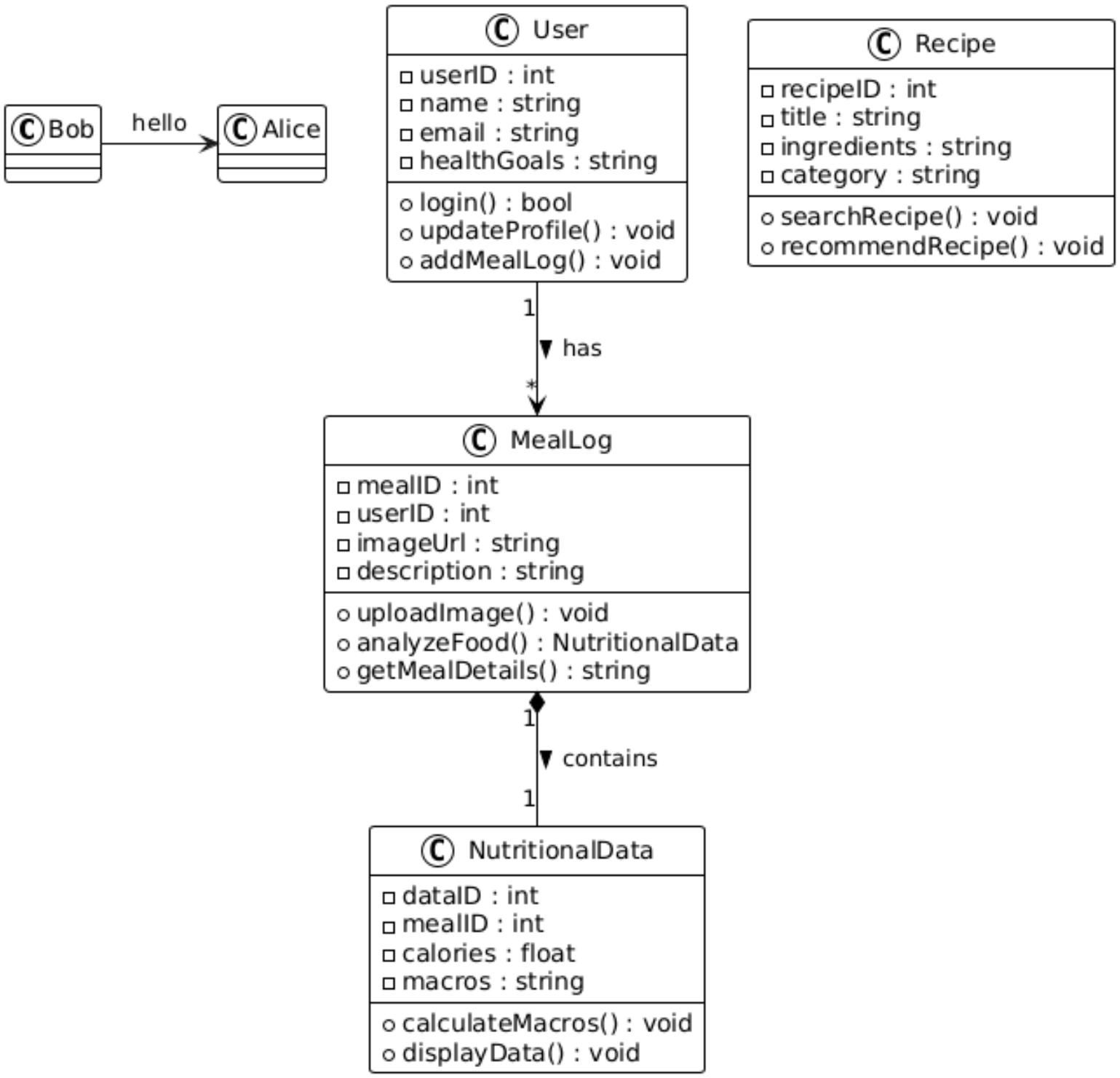

Bob
Alice
hello
User
UserID : int «PK»
Name : string
Email : string
HealthGoals : string
Recipe
RecipeID : int «PK»
Title : string
Ingredients : string
Category : string
1 to Many
MealLog
MealID : int «PK»
UserID : int «FK»
ImageUrl : string
Description : string
1 to 1
NutritionalData
DataID : int «PK»
MealID : int «FK»
Calories : float
Macros : string


Bob
hello
Alice
User
- userID : int
- name : string
- email : string
- healthGoals : string
+ login() : bool
+ updateProfile() : void
+ addMealLog() : void
Recipe
- recipeID : int
- title : string
- ingredients : string
- category : string
+ searchRecipe() : void
+ recommendRecipe() : void
1
has
*
MealLog
- mealID : int
- userID : int
- imageUrl : string
- description : string
+ uploadImage() : void
+ analyzeFood() : NutritionalData
+ getMealDetails() : string
1
contains
1
NutritionalData
- dataID : int
- mealID : int
- calories : float
- macros : string
+ calculateMacros() : void
+ displayData() : void

Database Schema Documentation

Detailed Data Model & Entity Specifications

1. User Table

Column Name	Data Type	Constraints & Notes
UserID	INT	Primary Key, AUTO_INCREMENT
Name	VARCHAR(100)	NOT NULL
Email	VARCHAR(150)	NOT NULL, UNIQUE
HealthGoals	VARCHAR(255)	NULL

2. MealLog Table

Column Name	Data Type	Constraints & Notes
MealID	INT	Primary Key, AUTO_INCREMENT
UserID	INT	Foreign Key REFERENCES User(UserID)
ImageUrl	VARCHAR(255)	NULL
Description	TEXT	NULL

3. NutritionalData Table

Column Name	Data Type	Constraints & Notes
DataID	INT	Primary Key, AUTO_INCREMENT
MealID	INT	Foreign Key REFERENCES MealLog(MealID), UNIQUE
Calories	FLOAT	NOT NULL
Macros	VARCHAR(255)	NULL

4. Recipe Table

Column Name	Data Type	Constraints & Notes
RecipeID	INT	Primary Key, AUTO_INCREMENT
Title	VARCHAR(150)	NOT NULL
Ingredients	TEXT	NOT NULL
Category	VARCHAR(50)	NULL

Relationships Note:

- 1 to Many: User (1) ——< MealLog (Many)
- 1 to 1: MealLog (1) ——— NutritionalData (1)